สมาชิก

กัญญาวี ศรีเหรา 673380026-6 รับผิดชอบ Part 1 และ Part 3 ทั้งหมด

รสริน เมืองหงษ์ 673380289-4 รับผิดชอบ Part 5 ทั้งหมด

สโรชา เสาทอง 673380296-7 รับผิดชอบ PART 2 และ Part 4 ทั้งหมด

ปวิศร์ แพงมา 673380047-8 รับผิดชอบ ASSESSMENT QUESTIONS

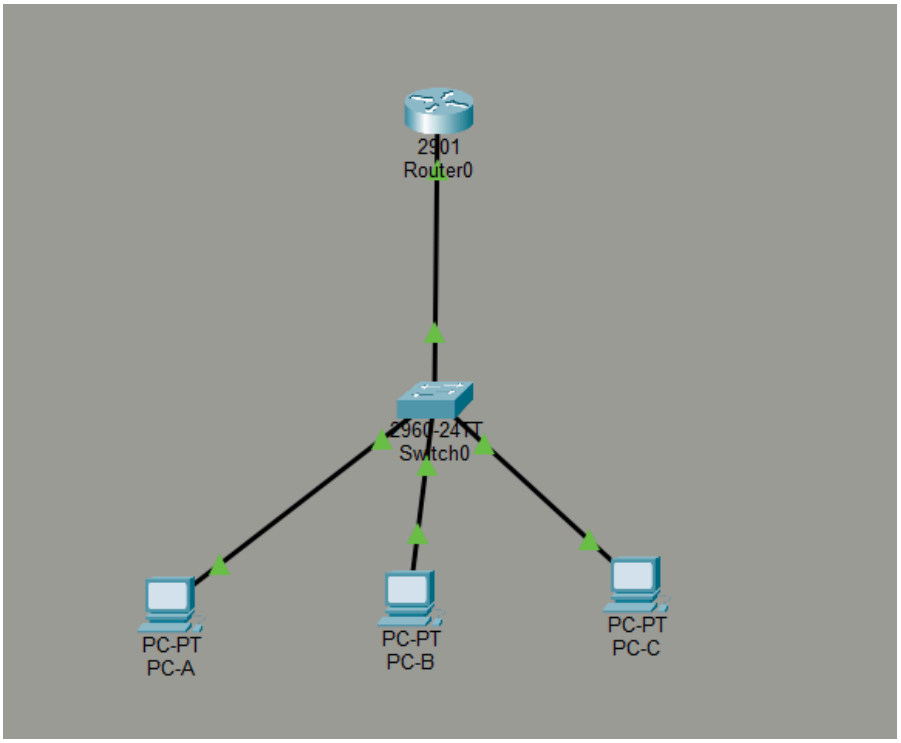
ธนกร ทองศรี 673380040-2 รับผิดชอบ worksheet ทั้งหมด

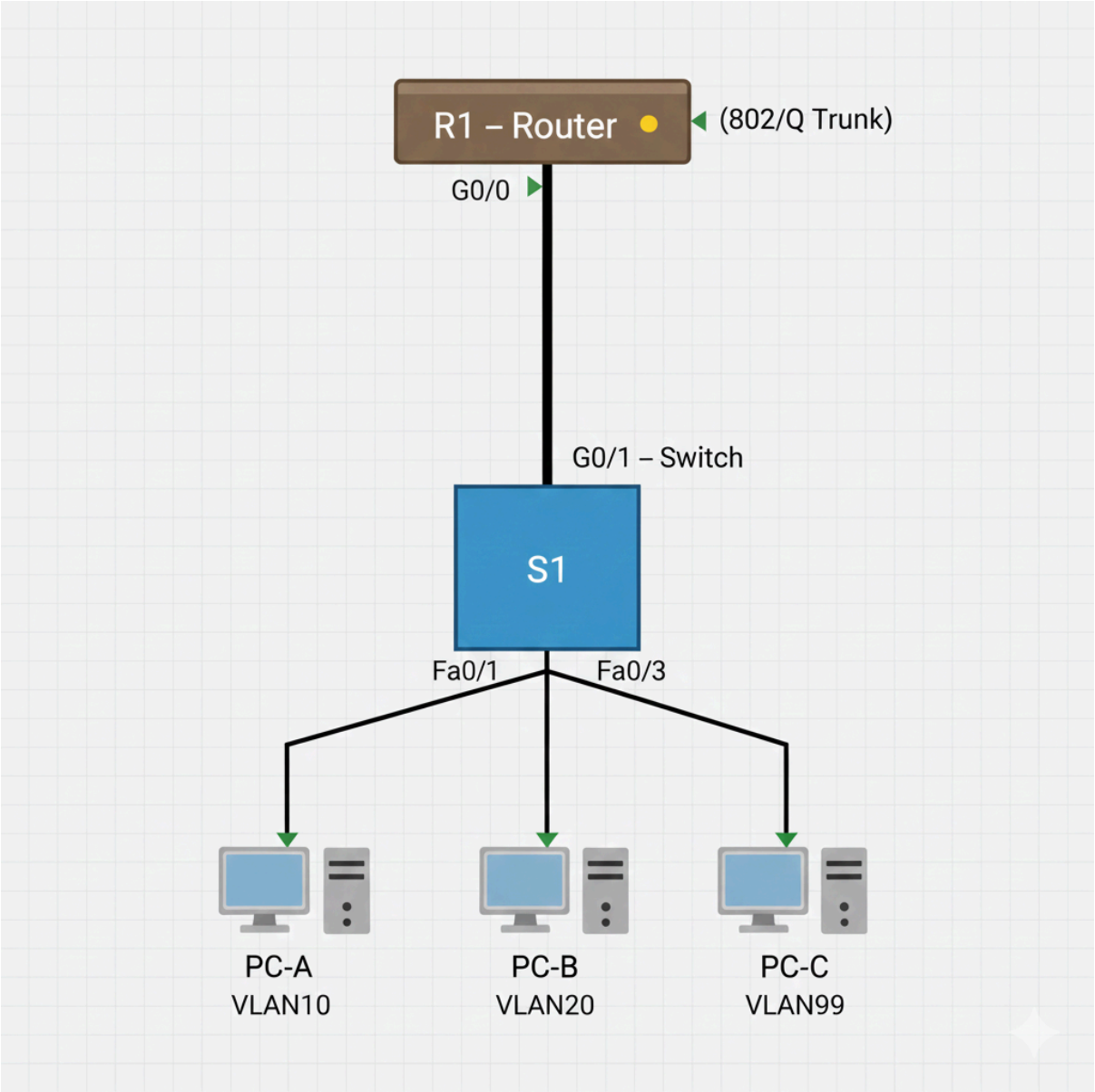
🧪 Lab 2: Secure & Scalable VLAN Design (Router-on-a-Stick)

📊 TOPOLOGY DIAGRAM

None

```
                    [ R1 – Router ]
                    G0/0 (802.1Q Trunk)
                              |
                              |
                     [ G0/1 – Switch ]
                    ┌──────────┴──────────┐
                    |         S1          |
                    |          |          |
            [Fa0/1]       [Fa0/2]       [Fa0/3]
                |          |           |
                |          |           |
                [PC-A] [PC-B]         [PC-C]
                VLAN10  VLAN20        VLAN99
```


2901
Router0
2960-24TT
Switch0
PC-PT
PC-A
PC-PT
PC-B
PC-PT
PC-C



📋 ADDRESSING TABLE

Network Summary

- **Major Network:** 192.168.10.0/24
- **Subnet Mask (/26):** 255.255.255.192
- **Subnets Used:**
  - VLAN 10 → 192.168.10.0/26
  - VLAN 20 → 192.168.10.64/26
  - VLAN 99 → 192.168.10.128/26

Device Addressing

Device	Interface	VLAN	IP Address	Subnet Mask	Default Gateway
R1	G0/0.10	10	192.168.10.1	255.255.255.192	N/A
R1	G0/0.20	20	192.168.10.65	255.255.255.192	N/A
R1	G0/0.99	99	192.168.10.129	255.255.255.192	N/A
S1	VLAN 99	99	192.168.10.131	255.255.255.192	192.168.10.129
PC-A	NIC	10	192.168.10.10	255.255.255.192	192.168.10.1
PC-B	NIC	20	192.168.10.70	255.255.255.192	192.168.10.65
PC-C	NIC	99	192.168.10.130	255.255.255.192	192.168.10.129

🎯 LAB OBJECTIVES

By completing this lab, you will be able to:

Part 1: Design and configure multiple VLANs on a switch **Part 2:** Implement Router-on-a-Stick for inter-VLAN routing **Part 3:** Apply subnetting using /26 networks **Part 4:** Isolate management and unused ports for security **Part 5:** Verify and troubleshoot Layer 2 and Layer 3 connectivity

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🔧 LAB INSTRUCTION STEPS

PART 1: SWITCH CONFIGURATION (35 minutes)

Step 1: Basic Switch Setup

```
None
Switch> enable
Switch# configure terminal
Switch(config)# hostname S1
```

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname S1
```

Step 2: Create VLANs

```
None
S1(config)# vlan 10
S1(config-vlan)# name USERS
S1(config-vlan)# exit

S1(config)# vlan 20
S1(config-vlan)# name SERVERS
S1(config-vlan)# exit

S1(config)# vlan 99
S1(config-vlan)# name MANAGEMENT
S1(config-vlan)# exit

S1(config)# vlan 999
S1(config-vlan)# name BLACKHOLE
S1(config-vlan)# exit
```

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname S1
S1(config)#vlan 10
S1(config-vlan)#name USERS
S1(config-vlan)#exit
S1(config)#vlan 20
S1(config-vlan)#name SERVERS
S1(config-vlan)#exit
S1(config)#vlan 99
S1(config-vlan)#name MANAGEMENT
S1(config-vlan)#exit
S1(config)#vlan999
               ^
% Invalid input detected at '^' marker.

S1(config)#vlan 999
S1(config-vlan)#name BLACKHOLE
S1(config-vlan)#exit
S1(config)#
```

Step 3: Secure Unused Ports (Security First)

```
None
S1(config)# interface range fa0/4-24
S1(config-if-range)# switchport mode access
S1(config-if-range)# switchport access vlan 999
S1(config-if-range)# shutdown
S1(config-if-range)# exit
```

```
S1(config)#interface range fa0/4-24
S1(config-if-range)#switchport mode access
S1(config-if-range)#switchport access vlan 999
S1(config-if-range)#shutdown

%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/6, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/7, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/8, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/9, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/10, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/11, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/12, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/13, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/14, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/15, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/16, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/17, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/18, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/19, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/20, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/21, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/22, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/23, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/24, changed state to administratively down
S1(config-if-range)#exit
```

Step 4: Assign Access Ports

```
None
S1(config)# interface fa0/1
S1(config-if)# switchport mode access
S1(config-if)# switchport access vlan 10
S1(config-if)# no shutdown
S1(config-if)# exit

S1(config)# interface fa0/2
S1(config-if)# switchport mode access
S1(config-if)# switchport access vlan 20
S1(config-if)# no shutdown
S1(config-if)# exit

S1(config)# interface fa0/3
S1(config-if)# switchport mode access
S1(config-if)# switchport access vlan 99
S1(config-if)# no shutdown
S1(config-if)# exit
```

```
S1(config-if-range)#exit
S1(config)#interface fa0/1
S1(config-if)#switchport mode access
S1(config-if)#switchport access vlan 10
S1(config-if)#no shutdown
S1(config-if)#exit
S1(config)#interface fa0/2
S1(config-if)#switchport mode access
S1(config-if)#switchport access vlan 20
S1(config-if)#no shutdown
S1(config-if)#exit
S1(config)#interface fa0/3
S1(config-if)#switchport mode access
S1(config-if)#switchport access vlan 99
S1(config-if)#no shutdown
S1(config-if)#exit
S1(config)#
```

Step 5: Configure Trunk to Router

```
None
S1(config)# interface g0/1
S1(config-if)# switchport mode trunk
S1(config-if)# switchport trunk allowed vlan 10,20,99
S1(config-if)# no shutdown
S1(config-if)# exit
```

```
S1(config)#interface g0/1
S1(config-if)#switchport mode trunk
S1(config-if)#switchport trunk allowed vlan 10,20,99
S1(config-if)#no shutdown
S1(config-if)#exit
S1(config)#
```

Step 6: Configure Management SVI

```
None
S1(config)# interface vlan 99
S1(config-if)# ip address 192.168.10.131 255.255.255.192
S1(config-if)# no shutdown
S1(config-if)# exit

S1(config)# ip default-gateway 192.168.10.129
S1(config)# end
S1# copy running-config startup-config
```

```
S1(config)#interface vlan 99
S1(config-if)#ip address 192.168.10.131 255.255.255.192
S1(config-if)#no shutdown
S1(config-if)#exit

S1(config)#ip default-gateway 192.168.10.129
S1(config)#end
S1#
%SYS-5-CONFIG_I: Configured from console by console

S1#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
```

PART 2: ROUTER CONFIGURATION (30 minutes)

Step 7: Basic Router Setup

```
None
Router> enable
Router# configure terminal
Router(config)# hostname R1
```

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R1
R1(config)#
```

Step 8: Enable Physical Interface

```
None
R1(config)# interface g0/0
R1(config-if)# no shutdown
R1(config-if)# exit
```

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R1
R1(config)#interface g0/0
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

R1(config-if)#exit
R1(config)#
```

Step 9: Configure Router-on-a-Stick Subinterfaces

```
None
R1(config)# interface g0/0.10
R1(config-if)# encapsulation dot1Q 10
R1(config-if)# ip address 192.168.10.1 255.255.255.192
R1(config-if)# exit

R1(config)# interface g0/0.20
R1(config-if)# encapsulation dot1Q 20
R1(config-if)# ip address 192.168.10.65 255.255.255.192
R1(config-if)# exit

R1(config)# interface g0/0.99
R1(config-if)# encapsulation dot1Q 99
R1(config-if)# ip address 192.168.10.129 255.255.255.192
R1(config-if)# exit
```

- `interface g0/0.X` → แยก VLAN
- `encapsulation dot1Q X` → บอกว่าเป็น VLAN อะไร
- `ip address` → Gateway ของ VLAN นั้น

Router ทำหน้าที่ **ให้ VLAN ต่างกันคุยกันได้**

```
None
R1(config)# end
R1# copy running-config startup-config
```

```
Router>ena
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R1
R1(config)#interface g0/0
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

R1(config-if)#exit
R1(config)#interface g0/0.10
R1(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.10, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up

R1(config-subif)#encapsulation dot1Q 10
R1(config-subif)#ip address 192.168.10.1 255.255.255.192
R1(config-subif)#exit
R1(config)#interface g0/0.20
R1(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.20, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up

R1(config-subif)#encapsulation dot1Q 20
R1(config-subif)#ip address 192.168.10.65 255.255.255.192
R1(config-subif)#exit
R1(config)#interface g0/0.99
R1(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.99, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.99, changed state to up

R1(config-subif)#encapsulation dot1Q 99
R1(config-subif)#ip address 192.168.10.129 255.255.255.192
R1(config-subif)#exit
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
```

PART 3: END DEVICE CONFIGURATION (15 minutes)

Configure each PC manually.

PC-A (VLAN 10)

- IP: 192.168.10.10
- Mask: 255.255.255.192
- Gateway: 192.168.10.1

PC-B (VLAN 20)

- IP: 192.168.10.70
- Mask: 255.255.255.192
- Gateway: 192.168.10.65

PC-C (VLAN 99)

- IP: 192.168.10.130
- Mask: 255.255.255.192
- Gateway: 192.168.10.129

PART 4: CONNECTIVITY VERIFICATION (20 minutes)

Step 10: Verification Commands

From S1

```
None
show vlan brief
show interfaces trunk
```

```
S1>show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Gig0/2
10   USERS                            active    Fa0/1
20   SERVERS                          active    Fa0/2
99   MANAGEMENT                       active    Fa0/3
999  BLACKHOLE                        active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                                Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

show vlan brief เป็นคำสั่งที่ใช้ แสดงข้อมูล VLAN ทั้งหมดบนสวิตช์ คือ

ส่วนประกอบ	ความหมาย
VLAN	หมายเลข VLAN
Name	ชื่อของ VLAN
Status	สถานะ (active = ใช้งานได้)
Ports	พอร์ตที่อยู่ใน VLAN นี้

```
S1>show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig0/1      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Gig0/1      10,20,99

Port        Vlans allowed and active in management domain
Gig0/1      10,20,99

Port        Vlans in spanning tree forwarding state and not pruned
Gig0/1      10,20,99

S1>sh
```

show trunk interfaces เป็นคำสั่งใช้ ตรวจสอบพอร์ตที่ทำงานแบบ Trunk บนสวิตช์ ว่า

- พอร์ตไหนเป็น trunk
- ใช้มาตรฐานอะไร (802.1Q)
- Native VLAN คืออะไร
- อนุญาต VLAN ไหนบ้างผ่าน trunk
- VLAN ไหน “ผ่านได้จริง” (ไม่ถูก block)

ใช้บ่อยในกรณีที่ ping ไม่ผ่าน

From R1

```
None
show ip interface brief
```

```
R1>show ip interface brief
Interface              IP-Address      OK? Method Status
Protocol
GigabitEthernet0/0     unassigned      YES unset  up
up
GigabitEthernet0/0.10  192.168.10.1    YES manual up
up
GigabitEthernet0/0.20  192.168.10.65   YES manual up
up
GigabitEthernet0/0.99  192.168.10.129  YES manual up
up
GigabitEthernet0/1     unassigned      YES unset  administratively down
down
Vlan1                  unassigned      YES unset  administratively down
```

show ip interface brief เป็นคำสั่งใช้ ดูสถานะ Interface ทั้งหมดของ Router (หรือ L3 Switch)

ส่วนประกอบ	ความหมาย
Interface	ชื่อพอร์ต
IP-Address	IP ที่กำหนด
OK?	IP ใช้ได้ไหม
Method	วิธีตั้ง IP
Status	สถานะทางกายภาพ
Protocol	สถานะทางตรรกะ

From PCs

```
None
ping 192.168.10.1
ping 192.168.10.70
ping 192.168.10.131
```

```
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time=1ms TTL=255
Reply from 192.168.10.1: bytes=32 time=1ms TTL=255
Reply from 192.168.10.1: bytes=32 time=1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.10.70

Pinging 192.168.10.70 with 32 bytes of data:

Reply from 192.168.10.70: bytes=32 time<1ms TTL=127
Reply from 192.168.10.70: bytes=32 time=1ms TTL=127
Reply from 192.168.10.70: bytes=32 time=1ms TTL=127
Reply from 192.168.10.70: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.10.70:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.10.131

Pinging 192.168.10.131 with 32 bytes of data:

Reply from 192.168.10.131: bytes=32 time<1ms TTL=255
Reply from 192.168.10.131: bytes=32 time<1ms TTL=255
Reply from 192.168.10.131: bytes=32 time<1ms TTL=255
Reply from 192.168.10.131: bytes=32 time=2ms TTL=255

Ping statistics for 192.168.10.131:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 0ms
```

Ping เป็นคำสั่งที่ใช้โปรโตคอลชื่อ ICMP (Internet Control Message Protocol)

เพื่อส่งข้อความทดสอบไปหาอุปกรณ์ปลายทาง

Reply from 192.168.10.70: bytes=32 time=2ms TTL=127
Reply from 192.168.10.70: bytes=32 time=1ms TTL=127
Reply from 192.168.10.70: bytes=32 time=1ms TTL=127
Reply from 192.168.10.70: bytes=32 time=1ms TTL=127
ถือว่า Ping ผ่าน

ความหมาย:

- Reply from ... ⟶ ปลายทางตอบกลับมาแล้ว
- มี time =วัดเวลาได้
- มี TTL = แพ็กเก็ตเดินทางได้จริง

Expected:

- PC-A ↔ PC-B ✅
- PC-A ↔ S1 Management IP ✅

PART 5: TROUBLESHOOTING CHALLENGE (25 minutes)

Scenario A: Wrong VLAN Assignment

- Move PC-B to the wrong VLAN
- Test connectivity
- Identify failure point
- Restore correct VLAN

1)ทำการย้าย vlan 20 ไปยัง 10

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface fastEthernet 0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

2)ทำการทดสอบ ping แต่ไม่สำเร็จ

```
C:\>ping 192.168.10.65

Pinging 192.168.10.65 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.10.65:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

3)เมื่อทำการปรับให้ vlan ให้เป็นเหมือนเดิม ทำให้สามารถ ping ได้

```
C:\>ping 192.168.10.65

Pinging 192.168.10.65 with 32 bytes of data:

Reply from 192.168.10.65: bytes=32 time<1ms TTL=255
Reply from 192.168.10.65: bytes=32 time<1ms TTL=255
Reply from 192.168.10.65: bytes=32 time<1ms TTL=255
Reply from 192.168.10.65: bytes=32 time=8ms TTL=255

Ping statistics for 192.168.10.65:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 8ms, Average = 2ms
```

Scenario B: Missing VLAN on Trunk

- Remove VLAN 99 from trunk
- Test access to S1
- Explain why management fails
- Restore configuration

1)ทำการลบ vlan 99 ออกจาก trunk

```
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface gigabitEthernet 0/1
Switch(config-if)#switchport trunk allowed vlan remove 99
Switch(config-if)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

2)ทำการ ping S1

```
C:\>ping 192.168.10.131

Pinging 192.168.10.131 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.10.131:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

3)Ping ไม่ได้เพราะ Management VLAN ไม่ถูกส่งผ่าน trunk ทำให้ PC กับ S1 อยู่คนละ VLAN กัน

4)ทำการเพิ่ม vlan 99 และทดสอบ ping อีกรอบ

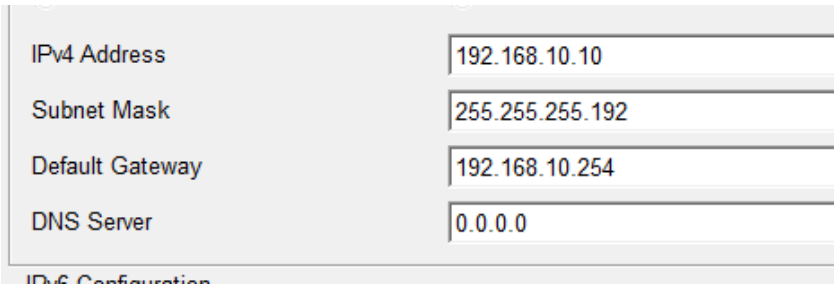
```
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#
Switch(config)#interface gigabitEthernet 0/1
Switch(config-if)#switchport trunk allowed vlan add 99
Switch(config-if)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

```
C:\>ping 192.168.10.131

Pinging 192.168.10.131 with 32 bytes of data:

Reply from 192.168.10.131: bytes=32 time<1ms TTL=254
Reply from 192.168.10.131: bytes=32 time<1ms TTL=254
Reply from 192.168.10.131: bytes=32 time<1ms TTL=254
Reply from 192.168.10.131: bytes=32 time<1ms TTL=254

Ping statistics for 192.168.10.131:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Scenario C: Wrong Gateway

- Change PC-A gateway
- Observe ping failure
- Use `ipconfig` and `ping`
- Fix gateway

1)ทำการแก้ Gateway ของ pc1 เป็น 192.168.10.254



2)ทดสอบ ping PC-C ซึ่งไม่สามารถ ping ได้

```
C:\>ping 192.168.10.130

Pinging 192.168.10.130 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.10.130:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

3)ใช้ ipconfic แล้วได้ผลลัพท์ดังนี้

```
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::260:5CFF:FE81:2384
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.10.10
   Subnet Mask.....................: 255.255.255.192
   Default Gateway.................: ::
                                     192.168.10.254

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0
```

4) แก้ gateway กลับมาเป็นเหมือนเดิม แล้วทดสอบ ping อีกครั้ง

```
C:\>ping 192.168.10.130

Pinging 192.168.10.130 with 32 bytes of data:

Reply from 192.168.10.130: bytes=32 time=7ms TTL=127
Reply from 192.168.10.130: bytes=32 time<1ms TTL=127
Reply from 192.168.10.130: bytes=32 time<1ms TTL=127
Reply from 192.168.10.130: bytes=32 time=8ms TTL=127

Ping statistics for 192.168.10.130:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 8ms, Average = 3ms
```

📝 LAB DELIVERABLES

Worksheet 1: Configuration Check

Checkpoint	Command	Expected	Actual	Pass/Fail
VLANs exist	show vlan brief	VLANs 10,20,99	VLANs 10,20,99	Pass
Trunk active	show int trunk	VLANs allowed	Port Vlans allowed on trunk Gig0/1 10,20,99	Pass
Router subifs	show ip int br	up/up	up/up	Pass
Inter-VLAN ping	ping	Success	Success	Pass

Worksheet 2: Design Understanding

Answer briefly:

1. Why is VLAN 99 used for management?
   ANS เพื่อแยกทราฟฟิกการจัดการออกจากทราฟฟิกของผู้ใช้ทำให้ปลอดภัยและเป็นระเบียบมากขึ้น
2. Why is VLAN 999 assigned to unused ports?
   ANS เพื่อป้องกันการเข้าถึงเครือข่ายโดยไม่ได้รับอนุญาตหากมีอุปกรณ์มาเสียบ พอร์ตนั้นจะไม่สามารถใช้งานเครือข่ายหลักได้
3. What does 802.1Q tagging do?
   ANS คือการเพิ่มแท็ก VLAN ลงในเฟรม Ethernet เพื่อให้สามารถส่งข้อมูลของหลาย VLAN ผ่าน ลิงก์ trunk เส้นเดียวได้
4. Why does each VLAN require a different subnet?
   ANS แต่ละ VLAN เป็น broadcast domain แยกจากกัน จึงต้องใช้ subnet ที่แตกต่างกัน เพื่อให้การทำงานของ Layer 3 routing ระบุและส่งข้อมูลระหว่าง VLAN ได้ถูกต้อง

🔍 VERIFICATION COMMANDS CHECKLIST

Router

- show run
- show ip route
- show arp

Switch

- show vlan brief
- show interfaces trunk
- show mac address-table

PCs

- ipconfig
- arp -a
- ping

🎓 ASSESSMENT QUESTIONS

1. Which device performs inter-VLAN routing?
   ANS Router0 ทำหน้าที่เป็นตัวส่งข้อมูลระหว่าง VLAN ต่างๆ
2. Why can PCs in different VLANs communicate?
   ANS Router0 ทำหน้าที่รับ packet จาก VLAN หนึ่ง ไปยัง VLAN อื่น ผ่าน Default Gateway ของแต่ละ VLAN
3. What breaks if the trunk is removed?
   ANS การสื่อสารระหว่าง VLAN จะล้มเหลว
4. Why is VLAN 1 avoided?
   ANS VLAN 1 เป็น VLAN ค่าเริ่มต้นของสวิตช์ และมักใช้สำหรับโปรโตคอลควบคุมเครือข่าย หากใช้ VLAN 1 จะมีความเสี่ยงด้านความปลอดภัยสูง
5. What security risk exists if unused ports remain active?
   ANS ผู้ไม่หวังดีสามารถเสียบอุปกรณ์เข้ามาและเข้าถึงเครือข่ายได้โดยไม่ได้รับอนุญาต อาจทำให้เกิดการโจมตีหรือดักข้อมูลได้

🎯 LAB COMPLETION CRITERIA

✔ All VLANs correctly created ✔ Router-on-a-Stick functioning ✔ Management VLAN isolated ✔ Unused ports secured ✔ All troubleshooting scenarios resolved

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